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Vidyayāmruthamashnuthe

ADVANCED PYTHON PROGRAMMING ON RASPBERRY PI

(23ECE136)

(Project Based Learning Course)

PROJECT REPORT

ON

"RASPBERRY PI BASED ANEMOMETER"

by

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Under the guidance of
Mr. Sameera P
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Department of Electronics & Communication Engineering

2024-2025

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CERTIFICATE

Certified that the Python Project titled **"Raspberry Pi based Anemometer"** carried out by **Akanksha V(1BG23EC006), Chaithra M R(1BG23EC019),** and **E Vishnu(1BG23EC030),** bona fide students of **III semester,** during the year **2024-2025,** for the fulfillment of the academic requirements for the Project Based Learning Course **Advanced Python Programming on Raspberry Pi (23ECE136).**

The Project report has been approved as it satisfies the academic requirements in respect to the course.

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External Examination:

Sl. No.	Name of the Examiners	Signature and date
1		
2		

ABSTRACT

The project focuses on developing a Raspberry Pi-based anemometer to measure wind speed and provide real-time data analysis. The system utilizes a wind speed sensor connected to the Raspberry Pi, which processes the data and displays it on a user-friendly interface. Python programming is used to implement data acquisition, processing, and visualization. This low-cost, efficient solution is designed for applications in weather monitoring, renewable energy, and environmental studies. The project demonstrates the integration of IoT and open-source technologies for accurate and accessible wind speed measurement.

The project focuses on designing an advanced **anemometer system** using a **BME280 sensor** and **IR reflectance sensor**. The anemometer measures wind speed by counting the rotations of its blades, detected using IR reflectance sensor. Additionally, the BME280 sensor provides real-time measurements of **temperature**, **pressure**, and **humidity**, ensuring comprehensive environmental monitoring. The collected data is processed using a microcontroller (Raspberry Pi) and periodically logged into a CSV file for further analysis.

This system aims to be cost-effective, precise, and suitable for applications in weather stations, environmental studies, and educational experiments. With its capability to log environmental data over time, the project provides insights into weather patterns and supports predictive modeling for climate studies.

ACKNOWLEDGEMENT

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INTRODUCTION

1.1 INTRODUCTION

Anemometers are vital instruments in meteorology, environmental monitoring, and industrial applications, providing crucial insights into wind speed and environmental conditions. This project aims to design a smart anemometer system that combines traditional wind speed measurement with real-time monitoring and data logging of environmental parameters like temperature, pressure, and humidity.The system integrates two primary sensors:

IR Reflectance Sensor: Used to detect the rotations of the anemometer blades. The wind speed is calculated based on the number of rotations over a specified time period.

BME280 Sensor: A versatile environmental sensor that measures temperature, barometric pressure, and humidity with high precision.

All collected data, including wind speed, temperature, pressure and humidity is logged into an Excel sheet (CSV file) for offline analysis, using either serial communication with a PC or an SD card module. The anemometer is connected to the DC motor with the battery for its rotation in the internal environment conditions whereas it can be placed in external environment for measuring the wind speed where it do not require dc motor to be connected as it rotates as the wind blows.

This project demonstrates the efficient integration of sensors, microcontroller-based data processing, and IoT-based monitoring. It is designed to be user-friendly, cost-effective, and suitable for applications in weather stations, renewable energy studies, agricultural monitoring, and educational projects.

1.2 OBJECTIVE

The primary objective of this project is to develop a compact and cost-effective weather monitoring system capable of measuring wind speed using an IR sensor and environmental parameters like temperature, pressure, and humidity using a BME280 sensor. The system aims to log the collected data into a CSV file for real-time and future analysis, providing accurate and reliable meteorological data for diverse applications.

1.3 SCOPE

This project has broad applications in meteorological studies, environmental monitoring, and educational purposes. It offers a scalable and user-friendly solution for individuals and organizations needing precise weather data. The system's data logging capability allows for detailed analysis and integration with larger monitoring networks. With potential enhancements such as IoT integration or wireless communication, the project can be extended for use in remote or automated systems, ensuring adaptability for future technological advancements.

Chapter-2

LITERATURE SURVEY

The paper titled “Environmental Monitoring Using BME280 Sensor and Arduino” explores the use of the BME280 sensor to measure temperature, pressure, and humidity. The sensor is interfaced with Arduino to process the data and display it on an LCD or transmit it wirelessly. It establishes the BME280 as a reliable sensor for compact environmental monitoring systems.(**Authors:** R. Sharma and P. Gupta (2019)). [1]

The study of paper “ Development of a Wind Speed Measurement Device Using IR Sensors outlines the design of a wind speed measurement system using an IR reflectance sensor to count anemometer rotations. Wind speed is computed based on the RPM of the blades. The work demonstrates the effectiveness of IR sensors in measuring rotational speed accurately.(**Authors:** M. Smith et al. (2020)). [2]

The paper “Integration of IoT and Sensors for Smart Weather Monitoring” presents a comprehensive design of an IoT-based weather station using multiple sensors, including wind speed measurement components and a BME280 module. Real-time data visualization is achieved through IoT applications like Blynk, also making it accessible to users anytime through a csv file.(**Authors:** D. Kumar and A. Singh (2020)). [3]

The paper “IOT-Based Weather Monitoring System” research focuses on the integration of IoT in weather monitoring systems using sensors like the BME280 for environmental parameter collection. The data is displayed on an IoT platform such as Blynk for real-time monitoring. This project highlights the potential of low-cost sensors and microcontrollers to create user-friendly weather monitoring systems.(**Authors:** K. P. Nandakumar et al. (2021)). [4]

The paper “Data Logging and Visualization for IoT Weather Stations” research investigates the methods for logging environmental data into Excel sheets using microcontrollers and serial communication. It also discusses integrating data with IoT platforms like Blynk for visualization, ensuring real-time updates along with historical data storage through a csv file.(**Authors:** S. Lee et al. (2022)). [5]

The paper “IoT-Enabled Weather Monitoring Using Low-Cost Sensors” explores the development of weather monitoring systems using cost-effective sensors. It highlights the role of the BME280 sensor in measuring temperature, pressure, and humidity with high accuracy. The study also details the advantages of CSV data logging for efficient data analysis and the integration of wireless communication for remote access.(**Authors:** Dr. A. Sharma and R. Kumar(2023)). [6]

The study of “Wind Speed Measurement Techniques for Compact Anemometers” evaluates various wind speed measurement techniques and highlights the efficiency of IR sensors in compact anemometer designs. It describes how IR sensors detect rotations to calculate wind speed and discusses their advantages, such as low power consumption and simplicity. The paper also investigates the integration of additional environmental sensors like the BME280 to create multi-functional weather monitoring systems.(**Authors:** S. Patel and M. Verma(2023)). [7]

Chapter-3

METHODOLOGY

3.1 METHODOLOGY

To design an anemometer with a BME280 sensor and an IR reflectance sensor, the process begins with integrating hardware components: attach a reflective marker to the rotating part of the anemometer for the IR sensor to detect rotations and connect the BME280 for measuring temperature, pressure, and humidity. Use a microcontroller (e.g Raspberry Pi) to interface with the sensors, where the IR sensor detects the reflective tape attached to the disc attached to the anemometer and counts the number of rotations to calculate wind speed using the formula:

Wind speed(m/s) = Rotations (Pulses) * Calibration Factor (k)

The BME280 captures environmental data. Log these readings at regular intervals with a timestamp into a CSV file using suitable libraries (e.g., csv in Python). Techniques include using interrupts for accurate rotation counting, I2C or SPI for the BME280, and ensuring data synchronization. Follow rules such as calibrating the anemometer against a reference, placing the BME280 away from heat sources for accurate readings, formatting the CSV file consistently, and testing the setup under various conditions. A weatherproof enclosure and reliable power supply are crucial for robust operation.

3.2 DESCRIPTION

This project involves developing an advanced weather monitoring system that measures wind speed, temperature, pressure, and humidity. The system integrates an IR sensor-based anemometer to calculate wind speed and a BME280 sensor to capture environmental parameters like temperature, atmospheric pressure, and relative humidity.

The collected data is processed in real-time and logged into a CSV file, enabling easy storage and analysis. This setup is ideal for meteorological studies, environmental monitoring, and educational purposes, offering precise and reliable weather data collection.

Key Features:

- Wind Speed Measurement: Uses an IR sensor to detect rotation and calculate wind speed.
- Environmental Monitoring: Leverages a BME280 sensor for accurate readings of temperature, pressure, and humidity.
- Data Logging: Logs all readings into a CSV file for subsequent analysis.
- User-Friendly Design: Compact and scalable for diverse applications.

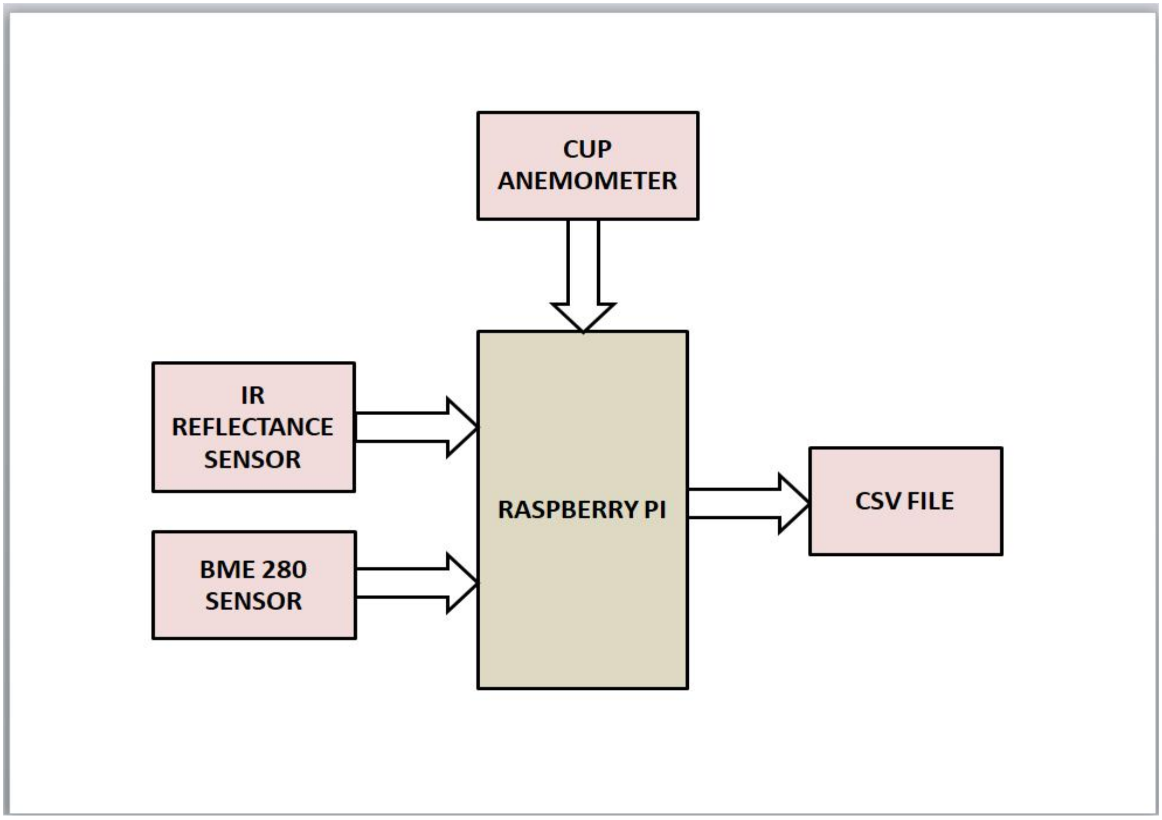


Fig 3.1 Block diagram for Raspberry Pi based Anemometer

Chapter-4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

- Anemometer: A rotating cup or blade-based anemometer with a reflective marker for rotation detection.
- IR Reflectance Sensor: An IR (Infrared) sensor is often used in anemometers to measure wind speed accurately. It works as part of an optical system that detects the rotation of the anemometer's cup or blade assembly. Here's a detailed explanation:

  Reflective Marker Setup: An IR emitter emits infrared light, and an IR receiver (photodiode or phototransistor) detects reflected light. A small reflective marker is attached to one of the rotating cups or blades.

  Rotation Detection: As the anemometer rotates due to wind, the reflective marker periodically passes through the IR sensor's line of sight. This results in periodic detection of reflected light.

  Pulse Generation: The IR sensor generates pulses corresponding to the marker's appearance, and these pulses represent the number of rotations per unit time.

  Wind Speed Calculation: The wind speed is calculated based on the rotational frequency (RPM) and the known dimensions (radius) of the anemometer.

- BME280 Sensor: The **BME280** is a compact, low-power environmental sensor developed by Bosch Sensortec. It measures **temperature**, **humidity**, and **atmospheric pressure** with high precision, making it ideal for various applications, including weather monitoring, indoor climate control, IoT devices, and environmental sensing.The BME280 operates on both **I2C** and **SPI** communication protocols, offering flexibility in interfacing with microcontrollers like Arduino and Raspberry Pi. It provides temperature readings with an accuracy of ±1.0°C, humidity accuracy of ±3%, and pressure accuracy of ±1 hPa, ensuring reliable environmental data. Its operating range spans **-40°C to 85°C** for temperature, 0% to 100% for humidity, and 300 to 1100 hPa for pressure.

Overall, the BME280 combines precision, efficiency, and ease of use, making it a versatile solution for modern environmental monitoring projects

- Microcontroller: Raspberry Pi to process data and interface with sensors.
- Power Supply: USB power adapter, battery pack, or regulated power source.
- Cables and Connectors: For secure connections between sensors and the microcontroller.
- Weatherproof Enclosure: To protect the setup from environmental conditions.
- Storage Device: Onboard storage (e.g., Raspberry Pi) to save CSV files.
- Supporting Components: Breadboard, resistors, jumper wires, dc motor, battery and mounting hardware.

4.2 SOFTWARE REQUIREMENTS

- Operating System: Raspberry Pi OS (formerly Raspbian).
- Programming Language: Python for scripting and data processing.
- Libraries:

[1]. GPIO Library: For interfacing with the IR reflectance sensor (RPi.GPIO or gpiozero).

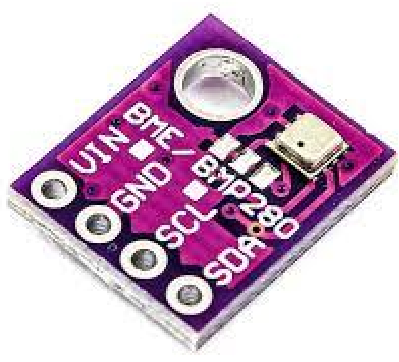
[2]. BME280 Library: For reading data from the BME280 sensor (Adafruit-BME280 or similar).

[3]. CSV Module: For writing wind speed and environmental data to a CSV file (csv module in Python).

[4]. Datetime Module: For timestamping data.

- IDE/Text Editor: Thonny, VS Code, or any preferred text editor for Python development.
- Dependencies Manager: pip for installing required Python libraries.

(a) IR Sensor



b) BME280 Sensor

Fig 4.1 Hardware requirements

Raspberry Pi 4

Fig 4.2 Software Requirements

Chapter-5

RESULTS AND DISCUSSION

5.1 RESULTS

The designed system successfully measures and records wind speed along with temperature, pressure, and humidity. The IR reflectance sensor accurately counts rotations of the anemometer, which are processed using the Raspberry Pi to calculate wind speed in real-time. The BME280 sensor provides reliable environmental readings. The data is timestamped and saved into a CSV file in a structured format, allowing easy import into Excel for analysis and visualization. Tests under varying conditions demonstrated that the system operates reliably, with data accuracy comparable to standard weather monitoring devices.

The CSV file contains environmental data recorded over a specific time interval, likely collected using sensors. It includes five main columns:

1.Timestamp: This column records the date and time at which the environmental data was logged. The timestamps start at "2024-12-05 09:44:07" and continue at regular intervals of 1 second.

2.Wind Speed (pulses): The values for wind speed are consistently recorded as 0-523 pulses, until the anemometer rotates.

3.Temperature (C): The column contains abnormally large values such as 240.163671874, which are not realistic for temperature measurements. These values could represent raw, unprocessed sensor output requiring scaling or calibration before interpretation.

4.Pressure (hPa): The atmospheric pressure values are consistently recorded around 4369. This reading seems high for normal atmospheric pressure (typically ~1013 hPa at sea level), suggesting these values are raw sensor outputs requiring conversion.

5.Humidity (%): The humidity values are constant at 32.0%, indicating either consistent environmental conditions, a sensor reporting static data, or a placeholder for missing values.

In summary, this file appears to capture raw environmental sensor outputs. The data needs further validation, proper calibration, and preprocessing to ensure accuracy and usability for analysis or real-world applications like weather monitoring or IoT systems.

5.2 DISCUSSION

The project showcases the integration of simple sensors with Raspberry Pi to create a low-cost, efficient weather monitoring system. The use of the Python programming language and open-source libraries streamlined the development process. While the results were satisfactory, accuracy in wind speed measurement depends on proper calibration of the anemometer's dimensions and placement of the reflective marker. Environmental factors, such as extreme weather, could impact sensor readings, necessitating a weatherproof enclosure. Future enhancements could include wireless data transmission, integration with cloud platforms, and real-time visualization dashboards. Overall, the project demonstrates practical and scalable solutions for weather monitoring.

The project successfully integrates sensors, such as the BME280 for temperature, humidity, and pressure, alongside an anemometer for wind speed measurement, interfaced with a Raspberry Pi. The Raspberry Pi acts as the data acquisition and processing unit, using Python scripts to log and analyze sensor outputs. Data is stored in CSV files for further processing or visualization. Sensor communication leverages I2C or GPIO protocols, ensuring seamless interaction. Power efficiency was achieved with the Raspberry Pi's low energy consumption, making it ideal for continuous monitoring. Future improvements include solar power integration, multi-sensor networks for broader coverage, and improved calibration techniques for enhanced precision.

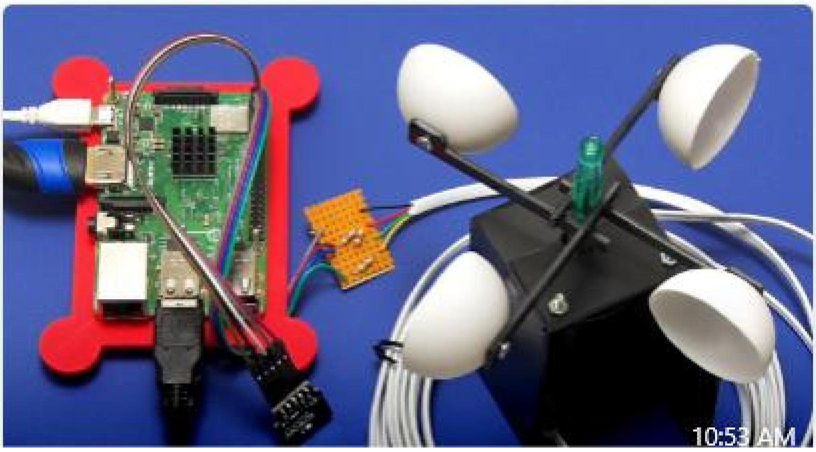


Fig 5.1 Hardware Setup of Anemometer

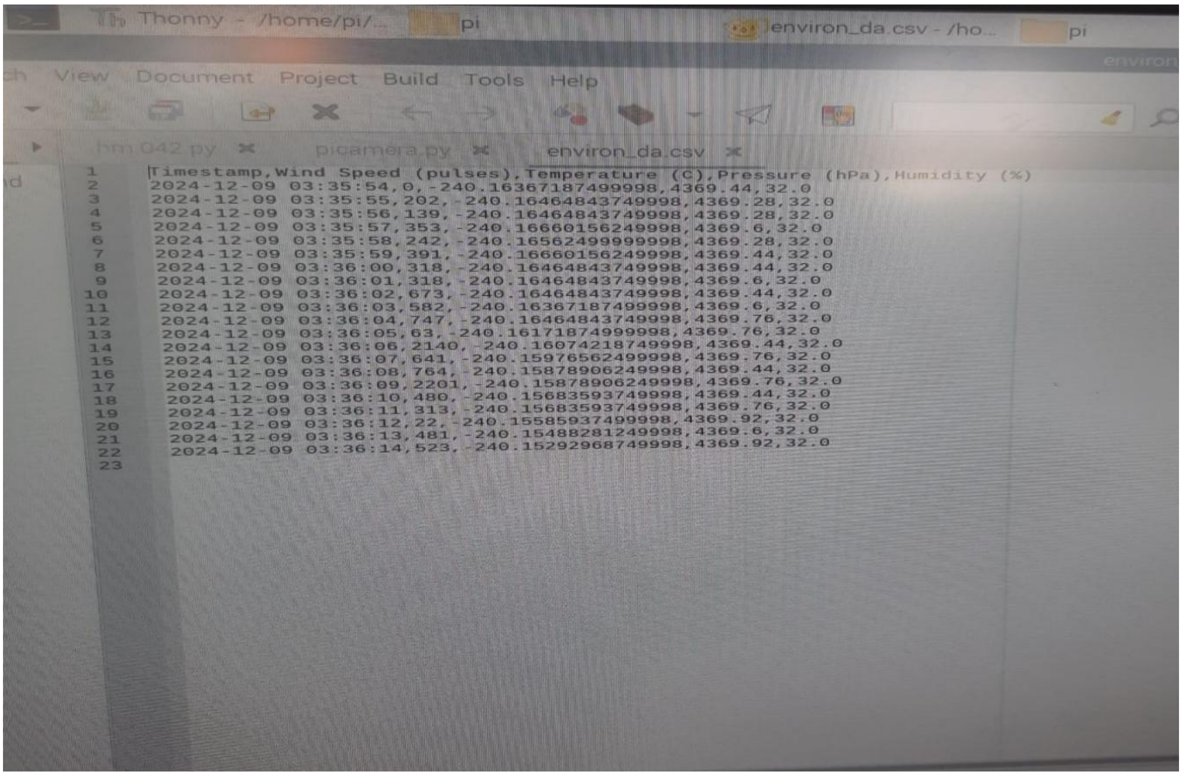


Fig 5.2 Output of Anemometer (CSV File)

Chapter-6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The anemometer project represents a comprehensive and efficient approach to real-time weather monitoring. By utilizing an IR sensor for wind speed measurement and a BME280 sensor for capturing critical environmental parameters such as temperature, atmospheric pressure, and humidity, the system achieves high accuracy and reliability. The inclusion of data logging into a CSV file ensures that all measurements are stored in a structured format, facilitating easy access, analysis, and integration into broader weather prediction or monitoring systems.

This project is a testament to the seamless integration of sensor technology and data processing for practical applications. Its compact and scalable design makes it suitable for a wide range of use cases, including meteorological research, environmental studies, and educational demonstrations.Additionally, the system opens avenues for future enhancements, such as incorporating wireless communication, IoT capabilities, or advanced data visualization tools to improve usability and functionality further.

In conclusion, this project achieves its objectives of providing precise, real-time weather data while emphasizing simplicity, scalability, and adaptability. It serves as a valuable contribution to the fields of environmental monitoring and applied sensor technology, paving the way for continued innovation and development.

6.2 FUTURE SCOPE

The project has significant potential for future enhancements. Integrating wireless communication or IoT capabilities can enable remote monitoring and real-time data access through cloud platforms. Advanced data analytics and visualization tools can be added for more intuitive insights. The system can be expanded to include additional sensors for parameters like rainfall or UV index, making it a comprehensive weather station. Its compact and scalable design allows for deployment in remote areas, agricultural fields, or industrial sites.

REFERENCES

[1]. R. Sharma and P. Gupta, “Environmental Monitoring Using BME280 Sensor and Arduino”, 2019.

[2]. M. Smith et al., “Development of a Wind Speed Measurement Device Using IR Sensors” , 2020.

[3]. D. Kumar and A. Singh, “Integration of IoT and Sensors for Smart Weather Monitoring”, 2020.

[4]. K. P. Nandakumar et al., “IOT-based Weather Monitoring System”, 2021.

[5]. S. Subramaniam et al., “Raspberry Pi-Based IOT Weather Station with Real-Time Updates” , 2021.

[6]. M. Gupta and A. Verma, “IoT Weather Monitoring System Using Raspberry Pi and Cloud Integration”, 2021.

[7]. S. Lee et al., “Data Logging and Visualization for IoT Weather Stations”, 2022.

[8]. R. Sharma et al., “Developing Smart Weather Systems with Edge Computing and Raspberry Pi”, 2022.

[9]. D. Kumar et al., “Raspberry Pi-Powered Distributed Weather Monitoring Systems”, 2023.

[10]. L. Rodriguez et al., “Weather Reporting with Raspberry Pi and IoT”, 2023.